NORWELCG Randomisation

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Introduction

This document describes the randomisation procedure for the NORWELCG study. The randomisation is set up according to the Specification for Randomisation procedure dated 2025-01-17 and according to the latest version of the protocol (version 1.05 dated 2024-05-09). This procedure is set up by the study statistician Inge Christoffer Olsen, and will be executed by an independent statistician at the Clinical Trials Unit of OSlo University Hospital, Erica Ponzi by setting the new seed. The specifications and the randomisation procedure is reviewed by the Quality Control statistician Nina Gunnes before the randomisation is executed.

Setup

According to the specifications: According to numbers gathered by the Norwegian Institution of Public Health, the number of births and caesarean sections (CS) are given as

Hospital no	Hospital name	Number of births	Percentage CS
1	Ahus	4818	13.4
2	Bergen	4243	8.8
3	KrSand	1796	8.6
4	RH	2268	12.3
5	Ullevål	5659	10.5
6	SUS	3945	11.9
7	SØ	2942	13.1
8	Drammen	1752	12.3
9	StOlav	3580	11.7
10	Tromsø	1243	13.4

With a cut-off at 2900 for the number of births, and 12.3% for the CSs we get the following strata:

Hospital no	Hospital name	Birth strata	CS strata	Strata
1	Ahus	High	High	1
2	Bergen	High	Low	3
3	KrSand	Low	Low	4
4	RH	Low	High	2
5	Ullevål	High	Low	3
6	SUS	High	Low	3
7	SØ	High	High	1
8	Drammen	Low	Low	4
9	StOlav	High	Low	3
10	Tromsø	Low	High	2

The stepped-wedge design is then according to the specification in the protocol:

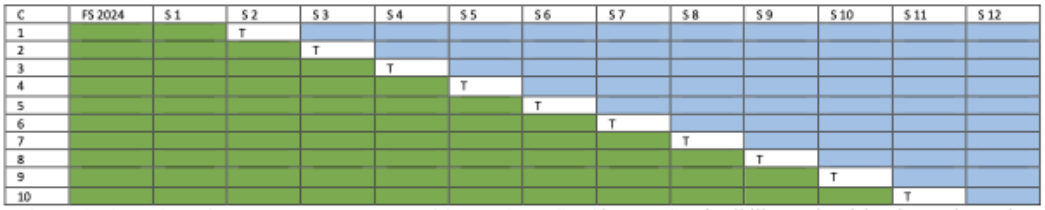


Figure 1. Overview of the study steps and transition periods. C = Cluster, FS = feasibility study of the electronic version of the LCG. S = steps of 5 weeks (S1-S12=60 weeks). Green colour denotes WHO partograph period, white colour denotes transition period for and teaching period, and blue colour denotes LCG period

The randomisation will then be performed as follows to ensure stratification by size and CS-rate:

1) Cluster 3 and 8 will be randomly assigned from strata 1 (hospital 1 or 7)
2) Cluster 4 and 7 will be randomly assigned from strata 4 (hospital 3 or 8)
3) Cluster 1, 2, 9 and 10 will be randomly assigned from strata 3 (hospital 2, 5, 6 or 9)
4) Cluster 5 and 6 will be randomly assigned from strata 2 (hospital 4 or 10)

This randomisation scheme will assure that there is good spread of the starting times according to important characteristics of the hospitals.

Randomisation

Following these specifications, the randomisation using seed 1234 produces the following randomisation:

Cluster	Hospital no	Hospital name	Birth rate	CS rate	Birth strata	CS strata	Strata
1	2	Bergen	4243	8.8	High	Low	3
2	9	StOlav	3580	11.7	High	Low	3
3	7	SØ	2942	13.1	High	High	1
4	8	Drammen	1752	12.3	Low	Low	4
5	10	Tromsø	1243	13.4	Low	High	2
6	4	RH	2268	12.3	Low	High	2
7	3	KrSand	1796	8.6	Low	Low	4
8	1	Ahus	4818	13.4	High	High	1
9	5	Ullevål	5659	10.5	High	Low	3
10	6	SUS	3945	11.9	High	Low	3